



# Assignment – Python List Slicing

 Duration: 20 minutes

 Topic: Working with Python Lists & Slicing

## Instructions

- Write each program in a separate Python file.
- Use proper variable names and comments.
- Display both input and output clearly.

### 1 Print the Last 4 Elements of a List

**Problem:** Given a list of numbers, display only the last 4 elements using slicing.

**Example Input:**

```
numbers = [5, 10, 15, 20, 25, 30, 35, 40]
```

**Expected Output:**

```
Last 4 elements: [25, 30, 35, 40]
```

### 2 Print Every 3rd Element

**Problem:** From a list of 12 elements, display every 3rd element using slicing with a step.

**Example Input:**

```
data = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
```

**Expected Output:**

```
Every 3rd element: [1, 4, 7, 10]
```

### 3 Reverse the List using Slicing

**Problem:** Write a program to print a reversed version of a list using slicing.

**Example Input:**

```
fruits = ["apple", "banana", "cherry", "dates"]
```

**Expected Output:**

```
Reversed list: ['dates', 'cherry', 'banana', 'apple']
```

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#### 4 Replace the Middle Elements of a List

**Problem:** Given a list of 6 elements, replace the middle two elements using slicing assignment.

**Example Input:**

```
colors = ["red", "green", "blue", "yellow", "orange", "purple"]
```

**Expected Output:**

```
After replacement: ['red', 'green', 'black', 'white', 'orange', 'purple']
```

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#### 5 Combine First and Last 3 Elements

**Problem:** Display only the first 3 and last 3 elements from a list and store them in a new list.

**Example Input:**

```
numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90]
```

**Expected Output:**

```
Combined list: [10, 20, 30, 70, 80, 90]
```